

Field Ion Electron Microscope (1951)

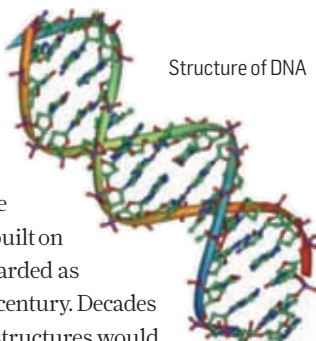
Erwin Mueller, a professor at Penn State University's department of physics, made an important contribution to Nanotechnology when he invented the field-ion electron microscope in 1951. For the first time in history, individual atoms and their arrangement on a surface could be seen. The device was a landmark advance in scientific instrumentation that allowed a magnification of more than 2 million times



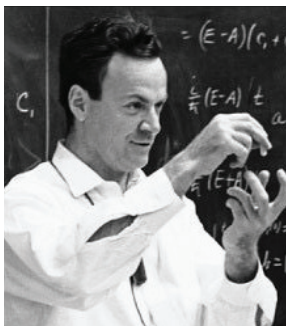
Field Ion Electron Microscope

Discovery of DNA (1953)

In 1953, Dr. James Watson and Professor Francis Crick published an article in *Nature* describing the double helix structure of DNA- deoxyribonucleic acid – which carried genetic information. They showed that when cells divide, the two strands that make up the DNA helix separate and a new “other half” is built on each strand, a copy of the one before. It is regarded as one of the landmark achievements of the 20th century. Decades later, DNA's ability to self-assemble into tiny structures would inspire researchers to use the same principles to develop nano-scale structures with specific dimensions and chemical properties.



Structure of DNA



Richard Feynman at Caltech

Richard Feynman (1959)

On December 29, 1959 Richard Feynman of the California Institute of Technology, gave what is considered to be the first lecture on technology and engineering at the atomic scale, “There’s Plenty of Room at the Bottom” at an American Physical Society meeting at Caltech. In his speech he stated, “What I want to talk about is the problem of manipulating and controlling things on a small scale. In the year 2000, when they look back at this